

CESM working log by Li Zhuo (lzhuo@albany.edu)

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1. Log into the server “head.its.albany.edu”

2. Clean the .bashrc file if you use Bourne-type shell, make sure you have a clean environment, and then source the environmental settings by Kevin:

```
source /network/rit/lab/snowclus/modules/intel24.bash
```

```
source /network/rit/lab/snowclus/bin/conda_python_only.bash
```

!! Remind: make sure all the old environment settings are cleared and removed before source new environment settings **!!**

3. Download CESM2 source code, follow the instruction on:

<https://github.com/ESCOMP/CESM/tree/release-cesm2.1.5>

3.1. First, cd into a directory (in my case, the directory is “/network/rit/lab/zhoulab_rit/lzhuo/”), clone the repository of CESM2:

```
git clone https://github.com/escomp/cesm.git my_cesm_sandbox
```

3.2. see what tags (versions) are available:

```
cd my_cesm_sandbox
```

```
git tag
```

3.3. checkout the version you want (I choose cesm2.1.5):

```
git checkout release-cesm2.1.5
```

3.4. run the checkout_external script to get the full source code and infrastructure:

```
./manageExternals/checkoutExternals (wait for several minutes for it to be finished)
```

4. Modify the machine and compiler configuration of CESM2 for burst-daes:

4.1. First, go to the directory where the corresponding files are located:

```
cd my_cesm_sandbox/cime/config/cesm/machines
```

4.2. Modify the file “config_machines.xml”:

Find my modification from lines 134-161 (according to Kevin’s file) in:

```
“/network/rit/lab/zhoulab_rit/lzhuo/my_cesm_sandbox/cime/config/cesm/machines/config_machines.xml”
```

The explanation for all these parameters can be found in:

https://esmci.github.io/cime/versions/master/html/users_guide/machine.html#config-machines-xml-machine-specific-file

4.3. Modify the file “config_compilers.xml”:

Find my modification from lines 1001-1052 according to Kevin’s file in:

```
“/network/rit/lab/zhoulab_rit/lzhuo/my_cesm_sandbox/cime/config/cesm/machines/config_compilers.xml”
```

4.4. Modify the file “config_batch.xml”:

Find my modification from lines 288-304 according to Peter’s file in:

```
“/network/rit/lab/zhoulab_rit/lzhuo/my_cesm_sandbox/cime/config/cesm/machines/config_batch.xml”
```

5. Create, setup, build and run the case

5.1. cd into the directory where “create_newcase” script is located and create the newcase

```
cd /network/rit/lab/zhoulab_rit/lzhuo/my_cesm_sandbox/cime/scripts/
```

```
/network/rit/lab/zhoulab_rit/lzhuo/my_cesm_sandbox/cime/scripts/create_newcase --compset FHIST_DARTC6 --res f19_f19 --machine burst-daes --run-unsupported --case TEST
```

FHIST_DARTC6 is the compset name, see the full compset list in:

<https://docs.cesm.ucar.edu/models/cesm2/config/compsets.html>

Further details about each component can be found at:

https://escomp.github.io/CESM/versions/cesm2.1/html/cesm_configurations.html#cesm2-components

f19_19 is the grid resolution, see the full resolution list in:

<https://www2.cesm.ucar.edu/models/cesm2/cesm/grids.html>

Further information about the grids:

https://escomp.github.io/CESM/versions/cesm2.1/html/cesm_configurations.html#cesm2-grids

TEST is the case name, burst-daes is the machine on which you will run CESM2.

Documentation for all the files after you create the case:

https://esmci.github.io/cime/versions/master/html/users_guide/create-a-case.html#results-of-calling-create-newcase

5.2. Change the PE layout

For details on PE layout:

https://esmci.github.io/cime/versions/master/html/users_guide/pes-threads.html

After make changes, use the **pelayout** tool to preview the PE layout:

```
./pelayout
```

The optimal PE layout differs from case to case. You need to try different PE layouts to find the optimal PE layout for your case. But keep in mind that **you cannot use over 4 nodes on burst-daes at a time**, other users still need nodes. (heard some professor used over 10 nodes previously to run the model on snow, leaving no nodes for others)

!! Remind: this step must be done before calling **case.setup**, if you modify the PE layout after calling **case.setup**, then you must run **case.setup --reset** and then call **case.build**. **case.build** will only recompile any source code that depends on values in **env_mach_pes.xml** **!!**

5.3. Set up the case

cd the directory of the case (TEST) and then set it up:

```
cd my_cesm_sandbox/cime/scripts/TEST
```

case.setup

5.4. Customizing your input variables

All namelists should be found at:

<https://docs.cesm.ucar.edu/models/cesm2/settings/current/>

To customize the driver input variables, use xmlchange command, for example:

```
./xmlchange RUN_STARTDATE=1850-09-01
```

However, if you want to modify the namelist of a specific component (for example, CAM6), you need to add the namelist to the corresponding user_nl_* files manually.

5.5. Build the case

case.build

Note: If you add the modified code, you must first clean the build and rebuild from scratch:

```
./case.build --clean-all
```

Further details about the build of the case can be found at:

https://esmpi.github.io/cime/versions/master/html/users_guide/building-a-case.html#

5.6. If everything goes well, you are going to submit the job, however, it will be a good idea to first call **preview_run**:

preview_run

which will output the environment for your run along with the batch submit and mpirun commands. Then, submit the job:

case.submit

check the status of your job:

squeue

if the model runs successfully to the end, then you should find a line

“SUCCESSFUL TERMINATION OF CPL7-cesm” at the end of *.cpl.log

(Have fun! 😊)

(Personal comment: CAM6 is more expensive to run than CAM4 and CAM5 because CAM6 introduces more fancy and complicated parameterization schemes, but do these new expensive parameterization schemes really help improve the simulation of clouds and other large-scale systems involving strong deep intense convection, such as MJO or BSISO? I think that the horizontal and vertical resolution of CAM6 is still too coarse to resolve clouds. To resolve some extreme weather systems, such as tropical cyclones, horizontal resolution as fine as 4 km is needed. I think that CESM2 can be used safely to study the climate over area where clouds have small impacts, such as Sahara Desert (Even such extreme dry area like the Sahara Desert is still under influence from the effects of deep intense convection in the remote tropics). However, extreme caution must be raised when CESM2 is used to study something in which clouds, or convection plays a key role, such as cloud feedback on surface temperature, or MJO. Some interesting discussion on this can be found in this paper by Kerry Emanuel:

<https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2019AV000129>)